

Cyprus AI — Architecture & API Cost Breakdown

What we use from OpenAI, what runs in-house, and why we built our own RAG

1. Overview

Cyprus AI is an internal company chatbot that answers employee questions from uploaded policy PDF documents. It combines our own in-house Retrieval-Augmented Generation (RAG) pipeline with OpenAI's large language model for the final answer generation. The vector search, document processing, and knowledge retrieval are built and hosted entirely by us — we pay OpenAI only for the text generation (the 'brain' that writes the answers).

2. What We Pay For (OpenAI)

We call the OpenAI Chat Completions API using the **gpt-5.4-mini** model. This is the only paid component. It acts as the language 'brain' and is used for these tasks:

- **Intent routing** — understands whether the user is greeting, asking about a document, or making small talk.
- **Query rewriting** — turns the user's question into optimized search terms.
- **Relevance evaluation** — judges whether the retrieved document chunks actually answer the question.
- **Answer generation** — writes the final natural-language answer using the retrieved facts.
- **Policy memory extraction** — when a document is uploaded, summarizes it and extracts key rules.

2.1 Typical Cost per Chat Message

One user question can trigger several small model calls. Here is a typical breakdown:

Step	Model Call	~Tokens In/Out	Est. Cost
1. Intent routing	gpt-5.4-mini	300 / 20	\$0.00005
2. Query rewriting	gpt-5.4-mini	300 / 200	\$0.0001
3. Relevance evaluation	gpt-5.4-mini	1,500 / 150	\$0.0003
4. Answer generation	gpt-5.4-mini	3,000 / 1,000	\$0.0008
5. Occasional retry	gpt-5.4-mini	1,500 / 256	\$0.0003
	TOTAL (typical)		≈ \$0.0015 / message

So a single Q&A costs roughly one-tenth of a US cent. At, say, 1,000 questions a month, the total OpenAI bill is about \$1–3 USD per month per user equivalent. Prices scale only with usage — there is no fixed license or server cost for the model.

3. What We Build & Host Ourselves (In-House)

Everything except the final text generation is built, owned, and operated internally. This keeps our documents private and our ongoing cost near zero for retrieval.

3.1 The RAG Pipeline (Built by us)

- **PDF Processing** — PyMuPDF4LLM, pdfplumber, and Tesseract OCR extract text, tables, and scanned pages from uploaded policy PDFs.

- **Chunking** — text is split into page-aware, searchable chunks (custom chunker using LangChain's splitter).
- **Embeddings (local, free)** — Ollama's nomic-embed-text converts each chunk into a 768-dimension vector. **Runs on our own server, no API cost.**
- **Vector Store** — ChromaDB stores the vectors and does fast semantic (cosine) search, plus our own keyword boost and cross-encoder re-ranking for higher accuracy.
- **Policy Memory** — when a document is uploaded, we distill it into a summary + key facts stored in PostgreSQL, injected at query time.
- **Conversation Memory** — multi-turn context stored in PostgreSQL with a 24-hour expiry.

3.2 Infrastructure

- **Application** — Python / FastAPI web service and admin portal.
- **Database** — PostgreSQL for users, folders, documents, conversations, and policy memory.
- **Vector DB** — ChromaDB (persistent) for embeddings.
- **LLM runtime** — Ollama runs locally for embeddings only.

4. Why We Don't Use OpenAI's Assisted/RAG Features

OpenAI offers its own 'Assistants' or file-search / retrieval features. We deliberately built our own RAG instead. The reasons:

- **Privacy & Control** — Company policy PDFs stay on our infrastructure. We never send raw documents to OpenAI; we only send the specific relevant text chunks needed to answer a single question.
- **Lower Cost** — Cloud vector search charges per query and per token on every retrieval. Our local ChromaDB search is free and instant, so we pay OpenAI only for the short final answer.
- **Freshness & Ownership** — We manage our own knowledge base, chunking, citation, and admin upload flow. Documents can be updated instantly through our portal, independent of any third-party tool.
- **Transparency & Auditing** — Every answer is backed by cited source chunks and a documented reasoning trace, giving HR/IT full visibility into what the bot used.
- **Accuracy Control** — We tune our own retrieval (keyword boost, re-ranking, relevance gates) to reduce hallucination and keep answers grounded strictly in our documents.
- **No Vendor Lock-in** — We can swap the model (OpenAI, or a local one) without rebuilding our retrieval system.

5. Summary

Component	Runs Where	Cost
LLM (gpt-5.4-mini)	OpenAI API	Per message (~\$0.0015)
Embeddings (nomic-embed-text)	Our server (Ollama)	Free
Vector Search (ChromaDB)	Our server	Free
Database (PostgreSQL)	Our server	Free
Application & Admin portal	Our server	Free

Bottom line: We keep our private data and the entire retrieval engine in-house (free), and pay OpenAI a few tenths of a cent for the model to write each answer. This gives us privacy, control, and dramatically lower running costs than using a fully cloud-hosted RAG service.

6. What Would It Cost to Move Fully to Azure AI Foundry?

A common question is whether we could consolidate everything onto Microsoft's Azure AI Foundry. The short answer: the LLM token cost would be roughly the same, but we would gain a standing monthly infrastructure bill — because Azure charges separately for the vector search, storage, networking, and monitoring that we currently run for free in-house.

6.1 Budget Scenario (50,000 questions per month)

The comparison below models a realistic enterprise workload — an internal knowledge assistant handling about 50,000 Q&A conversations per month, each using roughly 5,000 tokens of context plus a written answer.

Component	Our Hybrid (in-house)	Full Azure AI Foundry
LLM inference (gpt-4o-mini / gpt-4o)	~\$75 / mo	~\$75 – \$100 / mo
Vector search (ChromaDB vs Azure AI Search S1)	Free (local)	\$250 / mo
Document storage	Free (local disk)	~\$5 / mo
Private networking / endpoints	Free (local)	~\$40 – \$65 / mo
Monitoring / support overhead	~\$0 (minimal)	~\$30 – \$50 / mo
Hosting / compute overhead	Free (existing server)	~\$50 – \$100 / mo
	~\$50 – \$75 / mo	~\$450 – \$570 / mo

At this volume, moving fully to Azure AI Foundry costs roughly \$400–\$500 more per month (≈ \$5,000–6,000 more per year) than our current hybrid approach. The model itself is not the main driver — Azure AI Search (Standard S1) alone is about \$250/month, and networking, private endpoints, and monitoring add another ~\$150/month.

6.2 Yearly Comparison

Approach	Monthly	Yearly
Our hybrid (recommended)	~\$50 – \$75	~\$600 – \$900
Full Azure AI Foundry	~\$450 – \$570	~\$5,400 – \$6,800
Difference		~\$4,800 – \$5,900 saved / yr

6.3 Why the Jump?

- **Azure AI Search is a paid service** — it replaces our free in-house ChromaDB vector store. Standard S1 starts around \$250/month regardless of usage.
- **Infrastructure overhead** — private endpoints (~\$40–65/mo), monitoring, storage, compute, and support get billed separately instead of running on our own server.
- **Token prices are similar** — Azure's GPT-4o-mini is \$0.15/\$0.60 per million tokens, comparable to what we pay. The model is not the problem; the platform is.

Recommendation: Stay with the hybrid model. The paid LLM is the only thing worth buying per-token; hosting our own retrieval engine saves roughly \$400–\$500 every month while keeping our documents private and under our control.